A A S T U

Creativity and Innovation

Chapter 2

Processes (Steps) of Creativity and Innovation

Day 2 (Thursday) | Morning Session (2 hours) and Afternoon Session (2 hours)

Chapter overview

Yesterday we studied theories that explain where creativity comes from. Today we move from explanation to action. How does a creative idea actually travel from a vague problem to a working solution that benefits real people? Both creativity and innovation follow recognizable steps. When you know the steps, you stop waiting for inspiration and start managing it.

This chapter has two halves. In the morning we study the five steps of the **creativity process**: Preparation, Incubation, Illumination, Evaluation, and Implementation. In the afternoon we study the five steps of the **innovation process**: Ideation, Concept Development, Prototyping, Validation, and Deployment and Scaling. We end by looking at how creativity and innovation depend on each other, how organizations can strengthen them, and which challenges to expect.

One real work problem will travel with you through the whole day. You will choose it in the first hour and improve it step by step, so that by the end of the day you leave with a tested idea and a plan, and not only with notes.

Learning outcomes

By the end of Day 2, you will be able to:

- Describe the five steps of the creativity process and the five steps of the innovation process.
- Apply practical tools at each step, such as problem questions, incubation breaks, scoring matrices, SCAMPER, prototypes, and test plans.
- Explain how creativity and innovation depend on each other and how they differ.
- Identify ways organizations can strengthen creativity and innovation and the challenges that commonly block them.

- Develop one real workplace idea from problem to rollout plan and present it in two minutes.

Schedule for the day

Time block	Topic	Minutes
Morning	Welcome, Day 1 recap, and 2.1 Introduction to the creativity process	15
Morning	2.2.1 Preparation	20
Morning	2.2.2 Incubation	20
Morning	Break	10
Morning	2.2.3 Illumination	15
Morning	2.2.4 Evaluation	15
Morning	2.2.5 Implementation and morning recap	15
Afternoon	Welcome back, quick review, and 2.3 Introduction to the innovation process	15
Afternoon	2.3.1 Ideation and 2.3.2 Concept Development	30
Afternoon	2.3.3 Prototyping	15
Afternoon	Break	10
Afternoon	2.3.4 Validation and 2.3.5 Deployment and Scaling	25
Afternoon	2.4 Creativity and Innovation together, and two-minute pitches	25

How to use this module

- The blue **Key ideas** boxes summarize each step. The yellow **Activity** boxes are for group work, and the green **Check your understanding** boxes test yourself.
- The teal **Project thread** boxes build your own project. Keep your answers, because each step uses the work of the previous step.

- Use the running example of the student records office to see each step in action, then apply the same step to your own problem.

2.1 Processes (Steps) of Creativity

A **process** is a series of steps that lead to a result. We follow a process when we cook injera, register a student, or process a leave request. Creativity can seem mysterious, like a gift that arrives without warning. Yet studies of inventors, writers, and scientists show that creative work often passes through recognizable stages.

The best known description comes from the British social psychologist Graham Wallas, who in 1926 proposed four stages: preparation, incubation, illumination, and verification. This module follows his model but uses five steps that are more useful in organizations. We keep the first three and replace verification with two practical steps, **Evaluation** (is this idea good enough?) and **Implementation** (how do we put it to use?).



Real creative work is not a straight road. People often jump back, for example from Evaluation to Preparation when an idea fails and more information is needed. Treat the steps as a map and not as a strict timetable.

MORNING SESSION

Day 2 (Thursday) | 2 hours | The five steps of the creativity process

2.2 The Five Steps of the Creativity Process

2.2.1 Preparation

Preparation is the step where you **understand the problem and learn what is already known**. Many people skip it because they want to jump straight to solutions. This is a mistake. An answer to the wrong question has no value, however clever it is. Preparation includes defining the problem, collecting facts, observing how things really work, talking to the people affected, and learning from earlier attempts.

Useful tools

- **The "How might we" question.** Write the problem as an open question that invites many answers.
- **The five whys.** Ask "why" five times to move from a symptom to a root cause.
- **Observation and interviews.** Watch the process and ask users what is hard about it.
- **Facts and numbers.** Count, measure, and map the steps before you change them.

Hana in the records office. Hana first counts how many students arrive each hour and how long each request takes. She asks students why they wait. The five whys reveal that most students come on the last days of the semester because they were never told that clearance letters can be requested earlier. She writes her question: "How might we reduce the time students wait for a clearance letter from three hours to thirty minutes?"

Key ideas

- Preparation means understanding the problem before looking for solutions.
- A well-written "How might we" question guides the whole process.
- Facts, observation, and conversation with users are the raw material of later ideas.

Project thread 1: Choose and frame your problem (10 minutes)

Work in groups of three or four. Choose one real problem from your workplace that you can influence, such as long waiting times, lost documents, repeated errors, or confusing instructions. Then write:

- Your problem as a "How might we" question (include a target if possible).
- Three facts you still need and how you could find them.
- Two people you would ask, such as a client, a colleague, or a supervisor.

Check your understanding

1. Why is it risky to begin generating solutions before the problem is understood?
2. Turn this complaint into a "How might we" question: "Our files are always lost."

2.2.2 Incubation

After intense preparation, you may feel stuck. The next step is surprising. **Stop working on the problem for a while.** Incubation is the quiet period in which the mind continues to work in the background while you do something else. Many people report that solutions arrive in the shower, on a walk, in a taxi, or after a night of sleep. Research reviews suggest that a short break spent on an undemanding activity can help people find better solutions to open-ended problems.

Why might this work? During a break we stop repeating the same unsuccessful approach, weaker memories and loose connections have a chance to surface, and we return with a fresh view. Incubation is **not laziness and not procrastination**. It works only after real effort in Preparation, and it should have a time limit.

Ways to incubate

- Take a short walk or do light physical activity.
- Switch to a different, easy task for a short time.
- Sleep on it, and keep paper and pen near your bed.
- Talk about something unrelated, or talk to someone from a different field.
- Return to the problem at a planned time and note what new thoughts appeared.

Hana in the records office. After two days of data and interviews, Hana is tired of the problem. She decides to take a break and attends her cousin’s wedding preparations. At the wedding she notices how guests receive numbered seats and wait calmly because they know when it is their turn.

> **Key ideas**
>
> - Incubation is a planned pause that lets the mind work in the background.
> - It follows serious preparation and does not replace it.
> - Light activity, rest, and unrelated conversation are good incubators.

Activity 1: A Planned Pause (10 minutes)

Write your project question at the top of a page. Now stand up and take a five-minute silent walk, without discussing the problem. Return and write down any new thoughts that came up, even if they seem silly. Finally list three incubation habits you could use at work.

Check your understanding

1. Explain the difference between incubation and procrastination.
2. Which incubation activity works best for you, and why?

2.2.3 Illumination

Illumination is the moment of insight, often called the “aha moment”, when a new idea or connection suddenly appears. It feels like a light switching on. It is the most dramatic step, but it is also the most misunderstood. The flash does not appear from nothing. It is the visible tip of the work done in Preparation and Incubation.

Insights are often **small and fragile**. They can disappear within minutes if you do not write them down. Sometimes the first insight is only a piece of the solution, and we then shape it step by step, as the Honing Theory from Day 1 described.

How to welcome insight

- Keep a notebook or phone note always available.
- Write the idea at once in a few words and add a quick sketch.
- Do not judge the idea yet. Evaluation comes in the next step.
- Build on it. Ask what else could be combined with this idea.

Hana in the records office. On her way to work, Hana sees patients holding appointment cards at a health centre. She realizes that students could book a time slot to collect their letters, and that most requests could be prepared in advance. She writes it down immediately. During the day she adds two more ideas: an information board that tells students when to apply, and a simple online request form.

Key ideas

- Illumination is the sudden appearance of an idea after preparation and a pause.
- Insights are fragile, so capture them immediately.
- Do not judge ideas at this moment. Collect first and judge later.

Project thread 2: Idea burst (5 minutes)

Look again at your “How might we” question. Working alone, write as many ideas as you can in five minutes. Quantity matters more than quality at this moment. Then mark your three favourites with a star.

Check your understanding

1. Why should we avoid criticizing an idea at the moment it appears?
2. What was the source of Hana’s insight in the example?

2.2.4 Evaluation

Not every idea deserves to be implemented. Evaluation is the step where you **test ideas against clear criteria** and decide which are worth developing. This step connects to the Evolution Theory from Day 1, where selection decides which ideas survive. It also connects to the Systems Theory, because the people who will accept or reject the idea (the field) must be considered.

Common criteria include: Is it **useful** for the people we serve? Is it **new** enough to make a difference? Is it **feasible** with our skills and rules? Is it **affordable**? Will the people affected **accept** it? A simple scoring matrix makes the discussion clearer and fairer. Score each idea from 1 (weak) to 5 (strong) and add the scores.

Idea	Useful	New	Feasible	Afford-able	Accepted	Total
A. Appointment slots for letters	5	3	4	4	4	20

Idea	Useful	New	Feasible	Afford-able	Accepted	Total
B. Open two extra counters	3	1	3	2	4	13
C. Online request form	5	4	3	3	3	18

Hana’s group scores the ideas as above. The numbers do not make the decision for them. They **support a conversation**. The group decides that A is the strongest and that C can follow later, since the university does not yet have the required system. Notice that a low score on newness did not stop idea A, because usefulness and feasibility mattered more in this situation.

> **Key ideas**
>
> - Evaluation tests ideas against clear criteria such as useful, new, feasible, affordable, and accepted.
> - A scoring matrix supports discussion and does not replace judgment.
> - Consider the people who must accept the idea.

> **Project thread 3: Score your ideas (10 minutes)**
>
> Take your three starred ideas. Score each from 1 to 5 on the five criteria in the table above and add the totals. Choose the idea you will develop. Write one sentence explaining why you chose it.

> **Check your understanding**
>
> 1. Name three criteria you can use to evaluate an idea.
> 2. Why should the final decision not rest on scores alone?

2.2.5 Implementation

An idea that stays in a notebook has produced no change. Implementation is the step where you **take the first real actions** to put the idea to use. At this stage in the creativity process it

means starting small. You prepare a clear action plan, tell the people involved, and try the idea in a limited way. In the afternoon we will see how this grows into the formal innovation process.

Question	Your answer
What exactly will we do first?	
Who is responsible for each action?	
When will each action be done?	
What resources do we need?	
How will we know it is working?	

Expect resistance. People worry about extra work, about losing control, or about failure. Explain the benefits in simple words, listen to concerns, and invite colleagues to help improve the idea. **An idea that people help to shape is an idea that people help to defend.**

Hana in the records office. She asks her supervisor for permission to try appointment slots for one week with fifty students. She prepares a simple sign-up sheet and tells students about it through the notice board.

> **Project thread 4: Action plan (10 minutes)**
>
> Fill in the table above for your chosen idea. Be specific about who does what by when. Share your plan with another group and ask them to find one weakness.

> **Check your understanding**
>
> 1. Why is it wise to start the implementation in a small and limited way?
> 2. How can you reduce resistance from colleagues?

Morning recap (5 minutes)

The five steps of creativity are Preparation (understand the problem), Incubation (step back), Illumination (capture the insight), Evaluation (choose with criteria), and Implementation (start small and act). Before the afternoon, write the name of the person in your office whom you will ask for feedback on your project.

AFTERNOON SESSION

Day 2 (Thursday) | 2 hours | The five steps of the innovation process and how creativity and innovation work together

Welcome back: quick review (5 minutes)

Without looking at your notes, write the five steps of the creativity process in the correct order and give one tool for each. Compare with a partner.

2.3 Processes (Steps) of Innovation

In the morning we focused on how a person or a small team generates and shapes an idea. **Innovation goes further.** It is the full journey from idea to an improvement that real people use and value. The process is more organized, involves more people, needs resources, and depends on evidence. The steps below are used in many forms in design thinking and in product and service development. Their names may differ from one organization to another, but the logic is the same.

1 Ideation	2 Concept Development	3 Prototyping	4 Validation	5 Deployment and Scaling

The output of the creativity process becomes the input of the innovation process. Hana's tested idea of appointment slots now enters a more formal journey.

2.3.1 Ideation

Ideation is the **deliberate and structured generation of ideas**, usually in a group. It differs from a private flash of insight because the team plans it, uses tools, and builds on one another's thoughts. Ideation has two phases. First **widen** (produce many ideas), then **narrow** (select the strongest).

Alex Osborn, who invented brainstorming, proposed four rules: delay judgment, welcome wild ideas, aim for quantity, and build on the ideas of others. A helpful tool is **SCAMPER**, a list of seven questions that push us to change what already exists:

Letter	Question	Example for the records office
S	Substitute: what can we replace?	Replace paper request forms with a phone message
C	Combine: what can we join together?	Combine clearance and transcript requests in one visit
A	Adapt: what can we borrow from elsewhere?	Adapt the appointment cards of health centres
M	Modify: what can we enlarge, reduce, or change?	Shorten the form to one page
P	Put to another use: how else can it be used?	Use the waiting area as a help desk
E	Eliminate: what can we remove?	Remove the second signature for simple requests
R	Reverse: what if we turn it around?	Staff go to departments, not students to the office

> **Project thread 5: SCAMPER round (10 minutes)**
>
> Apply the seven SCAMPER questions to your chosen idea and write at least one answer for each. Then select the one or two changes that would make your idea stronger.

2.3.2 Concept Development

A promising idea is still only a sentence. Concept development **turns the idea into a clear description** that others can understand, discuss, and approve. A good concept answers who, what, why, and how, and it is written down in simple words on a single page. Writing it forces the team to find gaps.

Concept sheet	Your answer
Who is this for?	

Concept sheet	Your answer
What problem does it solve?	
How does it work, in three steps?	
What benefits will it bring?	
What resources and approvals are needed?	
What are the main risks?	

Hana's concept. For students who need clearance letters, the office will offer time slots of thirty minutes. Students sign up by phone or at the notice board, staff prepare letters before the slot, and students collect them within five minutes. The benefits are shorter waiting and calmer staff. It needs a shared schedule and the registrar's approval. The main risk is students who miss their slot.

> **Project thread 6: Write your concept sheet (10 minutes)**
>
> Complete the concept sheet above for your chosen idea. Keep each answer to one or two lines.

2.3.3 Prototyping

A prototype is a **quick, cheap, and rough version** of the idea that lets people see it, touch it, or try it. Its purpose is to learn, not to impress. The reasoning is simple. It is far cheaper to discover a mistake on paper than after the whole system has been built. This is sometimes called failing early and cheaply.

Kinds of prototypes

- **A sketch or storyboard** that draws the steps a user will follow.
- **A paper mock-up** of a form, sign, or screen.
- **A role play** in which colleagues act out the new process.
- **A minimum viable version**, the simplest working version that can test the main idea with real users.

Hana’s prototype. Hana does not wait for a computer system. She prints a sign-up sheet with time slots, places it at the office door, and offers the service for one week to fifty volunteer students. The tools cost almost nothing.

> **Activity 2: Five-minute prototype (10 minutes)**
>
> Draw a quick storyboard of your idea in four boxes. Box 1 shows the user with the problem, box 2 shows the user meeting your solution, box 3 shows how the solution works, and box 4 shows the result. Stick figures are welcome. Show it to a neighbor and ask what is unclear.

2.3.4 Validation

Validation asks a hard question: **does this really work for the people it is meant to help?** You test the prototype with real users, collect evidence, and compare the results with your goals. Do not rely only on your own opinion or on friendly colleagues who want to please you. Observe what users do and not only what they say.

Test plan question	Hana’s example
What do we want to learn?	Do time slots reduce waiting and do students use them?
Who will test it, and for how long?	Fifty students for one week
What will we measure?	Average waiting time, number of missed slots, student comments
What result means success?	Average waiting below thirty minutes and fewer than one in ten missed slots

After the trial, the team makes one of three decisions. It can **continue** with the idea, **adjust** it and test again, or **stop** and try a different idea. Stopping is not failure. It is a wise use of resources when the evidence is clear. In Hana’s case, waiting fell to twenty minutes, but some students missed their slot, so she adds a reminder message and continues.

> **Project thread 7: Test plan (5 minutes)**
>
> Fill in a four-row test plan like the one above for your own idea, with a clear success measure.

2.3.5 Deployment and Scaling

When the evidence is good, the idea is **deployed**, which means it is rolled out for regular use. **Scaling** means extending it to more people, more offices, or more locations. This step is where many good ideas fail, because the excitement of the pilot ends and the hard daily work begins. A careful rollout reduces the risk.

- **Name an owner** who is responsible for the new practice.
- **Train the people** who will use it and give them simple written guidance.
- **Communicate** why the change is happening and what each person must do.
- **Roll out in phases.** Move from one office to a few and then to all.
- **Support and monitor.** Collect feedback, fix problems quickly, and track the results.
- **Make it standard.** Update procedures, budgets, and job descriptions so that the change lasts.

Take care when scaling. An idea that worked in one office may need changes in another. The Shrinking Earth Theory from Day 1 reminds us to adapt and not only to copy. Hana’s office extends the slots to the transcript service in the second month and then shares the method with other university offices.

> **Key ideas**
>
> - Ideation widens and then narrows ideas, using tools such as brainstorming rules and SCAMPER.
> - A concept sheet describes the idea on one page, and a prototype lets people try it cheaply.
> - Validation tests with real users and ends in a decision to continue, adjust, or stop.
> - Deployment and scaling need an owner, training, communication, phased rollout, and monitoring.

Check your understanding

1. Why should a prototype be quick and rough?
2. What three decisions can follow a validation test?
3. Name three things that help an idea survive the scaling step.

2.4 Creativity and Innovation

We have studied the two processes separately. In real life they are closely connected. The final section looks at how they depend on each other, how they differ, and what organizations can do to support them or why they sometimes fail.

2.4.1 Interdependence of Creativity and Innovation

Creativity without innovation produces many ideas that nobody uses. Innovation without creativity repeats old solutions with new labels. They need each other. **Creativity supplies the ideas, and innovation turns them into value.** The relationship also runs in a loop. When an innovation is deployed, users give feedback and new problems appear, which start a new round of creativity. Hana's time slots, for example, soon led to a new question: how might we help students who have no phone?

2.4.2 Differences Between Creativity and Innovation

	Creativity	Innovation
Focus	Producing new and useful ideas	Putting ideas into practice to create value
Typical people	Individuals and small teams	Teams, managers, and whole organizations
Resources	Time, curiosity, and imagination	Time, money, approvals, and training
Risk	Low, since ideas cost little	Higher, since implementation costs resources

	Creativity	Innovation
Measure of success	Originality and usefulness of the idea	Real improvement and adoption by users

2.4.3 Enhancing Creativity and Innovation in Organizations

- **Leadership that listens.** Leaders show interest in new ideas and protect people who try them.
- **Psychological safety.** Amy Edmondson's research shows that teams learn more when members feel safe to speak up and admit mistakes.
- **Time and space.** Give staff time to think, experiment, and meet across departments.
- **Simple channels for ideas.** Suggestion systems, short idea meetings, and clear answers to every suggestion.
- **Diverse teams.** People of different ages, skills, and backgrounds bring different views.
- **Learning from failure.** Review pilots that did not work and share what was learned.
- **Recognition.** Thank people for effort and results, and not only for success.

2.4.4 Challenges in Creativity and Innovation

Challenge	Why it happens	A possible response
Resistance to change	People fear extra work or loss of status	Involve them early and explain the benefits
Fear of failure	Mistakes are punished or ridiculed	Run small pilots and praise learning
Limited resources	Budgets and time are tight	Start with cheap prototypes and use existing tools
Short-term pressure	Daily tasks crowd out improvement	Set aside protected time each week
Rigid hierarchy	Ideas must pass through many levels	Create safe, direct channels for suggestions

Challenge	Why it happens	A possible response
Weak follow-through	No owner or plan after the pilot	Name an owner and write a rollout plan

Activity 3: Two-Minute Project Pitch (20 minutes)

Each group presents its project thread in two minutes, using your worksheets from today. Cover these five points in order:

- The problem as a "How might we" question.
- The chosen idea and why it was chosen.
- The prototype and how it was tested.
- The plan to deploy and scale it.
- One challenge you expect and how you will respond.

Listeners give one positive comment and one helpful suggestion to each group.

Chapter summary

- Creativity and innovation both follow recognizable steps that can be learned and managed.
- The creativity process has five steps: Preparation, Incubation, Illumination, Evaluation, and Implementation.
- The innovation process has five steps: Ideation, Concept Development, Prototyping, Validation, and Deployment and Scaling.
- Creativity and innovation depend on each other, although they differ in focus, people, resources, and risk.
- Organizations strengthen them through listening leaders, psychological safety, time, diverse teams, and learning from failure.
- Common challenges include resistance, fear of failure, limited resources, and weak follow-through. Each has practical responses.

Key terms

Term	Meaning
Process	A series of steps that lead to a result
Preparation	Understanding the problem and gathering knowledge before looking for solutions
Incubation	A planned pause that lets the mind work in the background
Illumination	The moment of insight when a new idea appears
Prototype	A quick, cheap version of an idea used to learn and test
Validation	Testing an idea with real users and comparing the results with goals
Scaling	Extending a successful idea to more people, offices, or places
Psychological safety	A shared belief that it is safe to speak up and admit mistakes

Review questions for discussion or homework

1. Explain the five steps of the creativity process using an example from your own workplace.
2. Why is Preparation often the most neglected step, and what are the risks of skipping it?
3. Describe the difference between Illumination in the creativity process and Ideation in the innovation process.
4. Choose one prototype idea for a problem in your office and explain how you would test it.
5. How do creativity and innovation depend on each other? Give an example of each failing without the other.
6. What are three actions a manager can take to build psychological safety in a team?

Further reading

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Your next steps after training

Within the next thirty days, take your project thread to your supervisor and ask for permission to run your small trial. Share what you learned with at least one colleague. Meet again with your group after one month and compare results. Thank you for your effort and your ideas over these two days.